Eggshell–Agar Biocomposites

Development and characterisation for sustainable interior applications

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Background

Material basis

94–97%

calcium carbonate by mass

0.85–2.66

specific gravity, below most mineral fillers

William & Owen (1995) · Torres et al. (2010) · Owuamanam & Cree (2020)

Two routes



The case for eggshell as a mineral phase is settled. This paper is about the phase that binds it.

Prior applications

Established uses of eggshell waste in engineered materials

Application	Binding phase	Reference
Partial cement replacement in concrete	Portland clinker	Teara et al. (2019); Bhartiya & Dubey (2018)
Compressed laterite brick	Soil – lime	Adogla et al. (2016)
Hot-mix asphalt	Bitumen	Razzaq & Yousif (2017)
Clay stabilisation	Clay mineral	Soundara & Vilasini (2015)
Eco-ceramic wall tile	Ceramic body	Vilarinho et al. (2022)
Particulate filler in PVC	Poly(vinyl chloride)	Skórczewska et al. (2020)
Filler in flexible polyurethane foam	Polyurethane	Victor et al. (2019)
Ceramic–metal composite	Metal matrix	Bilici et al. (2023); Yönetken et al. (2016)
Bone tissue scaffold	Calcium phosphate	Shafiei et al. (2019)
Wallpaper	Cellulosic furnish	Tutuş et al. (2018)

The burden is carried by the binder, and the binder is left untouched.

MINERAL PHASE · recovered from waste

BINDING PHASE · cementitious, ceramic or petrochemical

Research gap

Three gaps follow from the preceding two slides

01

The binding phase is never biological

No study reports an eggshell composite in which both phases are of biological origin.

02

Agar is characterised only as an unfilled film

Never as the binder of a mineral-filled solid in bending.

03

Property sets are reported in fragments

Property data is scattered across separate studies on separate formulations.

This is not a novel constituent. It is a documented protocol and a single coherent property set.

Aim

To document a reproducible production protocol for a fully bio-based eggshell–agar composite, and to characterise it against published test standards.

Objectives

01	02	03	04
Document the protocol	**Characterise to standards**	**Position against two families**	**Derive an evaluation criterion**

The fourth objective is the one the paper turns on.

Standards applied: ASTM D790-17 · ISO 11925-2:2020 · ASTM D570-98(2018).

Scope

Within scope

Dimension	This study
Formulations	Five — three compressive, two flexural
Specimens	Three per formulation for mechanical testing
Properties	Compression, flexure, ignitability, water absorption, bulk density
Standards	ASTM D790-17 · ISO 11925-2:2020 · ASTM D570-98(2018)
Conditions	Ambient cure, single laboratory, single operator
Application	Non-loadbearing interior surfaces

Outside scope

- Durability and ageing
- Fire classification
- Life-cycle assessment
- Structural use

Property coverage

Compression | **Flexure** | **Ignitability** | **Water absorption** | **Bulk density** | Durability | Fire classification | Life-cycle | Structural use

The filled boxes were determined in this study. The outlined boxes were not.

Method

A material-driven sequence in four stages

1 → 2 → 3 → 4

1	2	3	4
Produce	**Measure**	**Position**	**Propose**
every parameter recorded	against published standards	against two material families	from the observed behaviour

The properties came first, and the application envelope was derived from them.

Material-driven design after Karana et al. (2015)

Research questions

01	**Which formulation variable governs compressive strength?**	Formulation
02	**How does plasticiser content affect flexural behaviour?**	Formulation
03	**How does the composite compare with conventional interior materials?**	Benchmark
04	**How does it compare with other bio-based composites?**	Benchmark
05	**Is compressive strength an adequate evaluation criterion for this class?**	Criterion

The first four are answered by measurement. The fifth is answered by argument.

Hypothesis

A fully bio-based eggshell–agar composite can be produced repeatably under ambient conditions, and its performance as an interior surface material is governed by deformation capacity rather than by compressive strength.

Two clauses, tested separately

Clause 1	**Producible repeatably under ambient conditions**	➡ tested on slides 10–13
Clause 2	**Governed by deformation capacity rather than strength**	➡ tested on slides 15–22

The dissertation behind this paper frames a broader hypothesis around spatial resilience; this is its materials-engineering component.

Stage 1 · Mineral phase

Progressive size reduction of the shell to a sieved powder



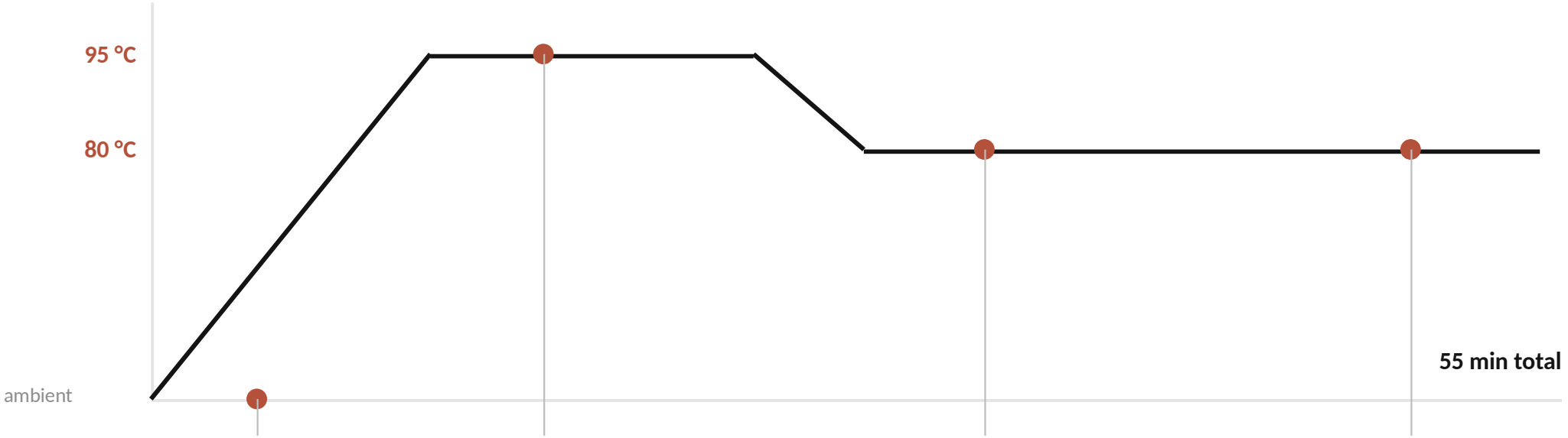
Whole shell	Cleaned	Heat-treated	Milled	Sieved
collected from a single bakery	membrane and residue removed manually	decontamination and drying	laboratory mechanical grinder	passing fraction retained
—	—	**120 °C, 20 min**	**to homogeneous powder**	**150 µm mesh**

The 150 µm fraction was selected so that the mineral phase would be comparable with the eggshell polymer-composite literature.

The heat step is decontamination and drying rather than sterilisation, because sterility was never verified.

Stage 2 · Binder preparation

Temperature–time profile of the 55-minute preparation



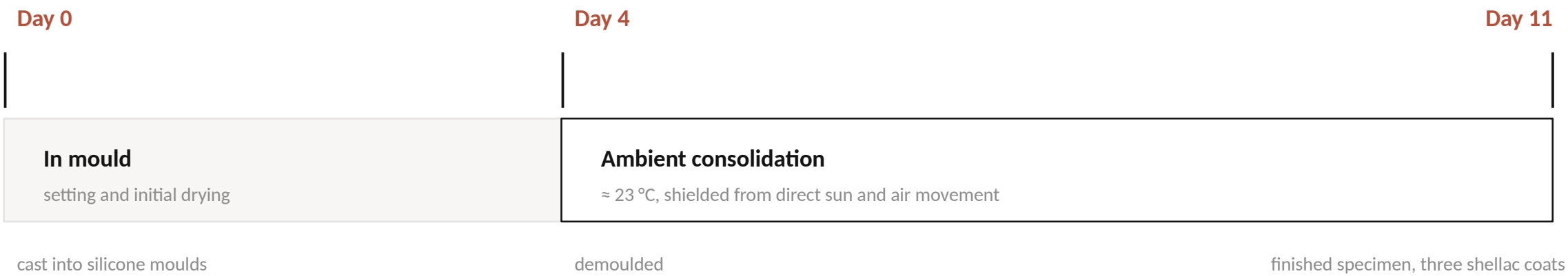
Agar is thermoreversible: it melts on heating and sets again on cooling, so no hardener and no cure temperature are needed.

The agar is added a little at a time under constant stirring, because added all at once it forms agglomerates that never redisperse.

Schematic. The set points and the total time are recorded; the ramp rates are not, and stirring was manual.

Stage 3 · Casting and cure

Eleven days from casting to finished specimen, without thermal input



After the 55-minute preparation the material receives no further heat, so the entire cure happens at room temperature.

Silicone moulds · 50 mm cube moulds for compression, flat moulds otherwise.

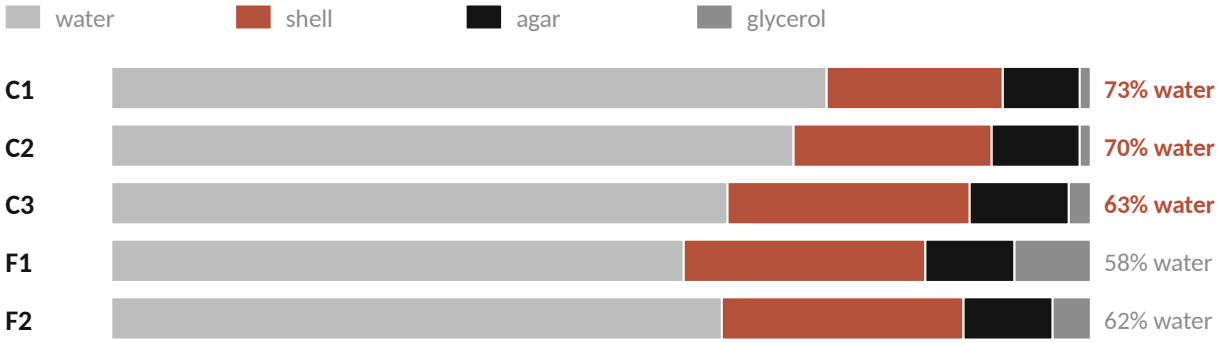
Limitation the composite shrinks as the water evaporates, and the dried dimensions were not logged, so every dimension reported here is a mould dimension.

Formulations

Five mixes, with the derived ratios that govern the results

ID	Test	Water g	Shell g	Agar g	Glycerol g	Shell : binder	Solids : water
C1	Compression	325	80	35	5	**2.00**	0.37
C2	Compression	310	90	40	5	**2.00**	0.44
C3	Compression	280	110	45	10	**2.00**	0.59
F1	Flexure	450	190	70	60	1.46	0.71
F2	Flexure	480	190	70	30	1.90	0.60

Batch composition by mass

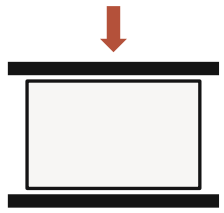


Solids concentration is the only variable, and the bars show it: the water band gets shorter from C1 to C3.

The glycerol was halved while the batch mass stayed the same, so the water rose to compensate.

Test programme

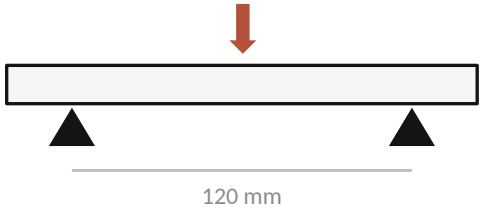
Four specimen geometries



Cube compression

50 mm cube, platen-loaded

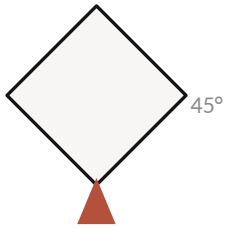
n = 3 per mix



Three-point flexure

180 × 60 × 20 mm, 120 mm span

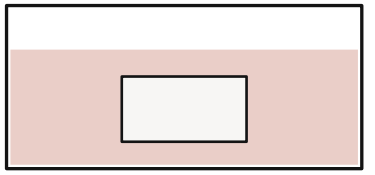
ASTM D790-17 · n = 3 per mix



Edge ignition

40 × 50 mm at 45°, 15 s flame

ISO 11925-2:2020 · n = 2



Immersion

40 × 50 mm, shellac-coated, 24 h

ASTM D570-98 · n = 1

The mechanical values are means of three specimens, and since the individual values were not retained, standard deviations cannot be reported.

The fire, absorption and density figures are single determinations.

All testing: Advanced Construction Materials Research Laboratory, Hacettepe University. · Schematic, not to scale.

Compression

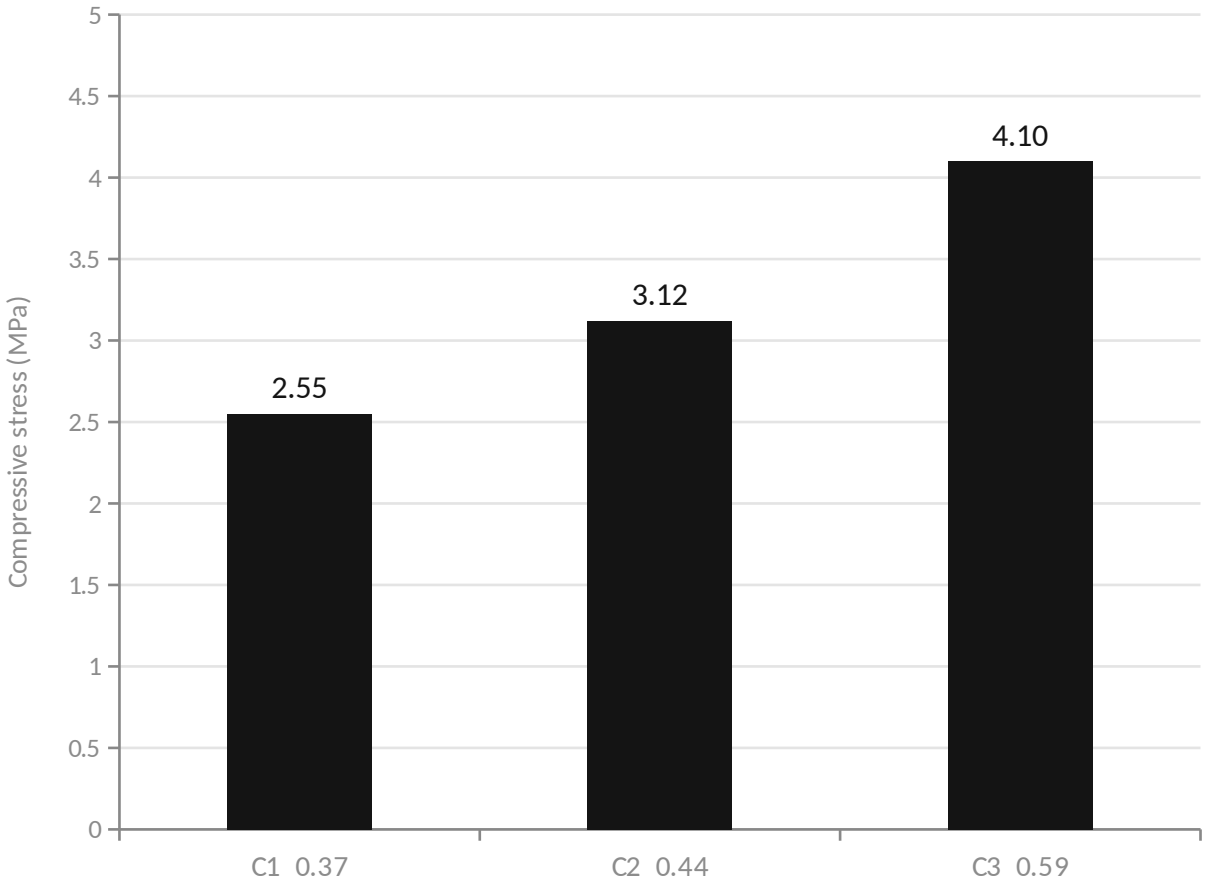
Measured values

ID	Solids : water	F_{max} kN	σc MPa
C1	0.37	3.89	2.55
C2	0.44	4.30	3.12
C3	0.59	**6.80**	**4.10**

Strength here is governed by packing density, not by mineral content.

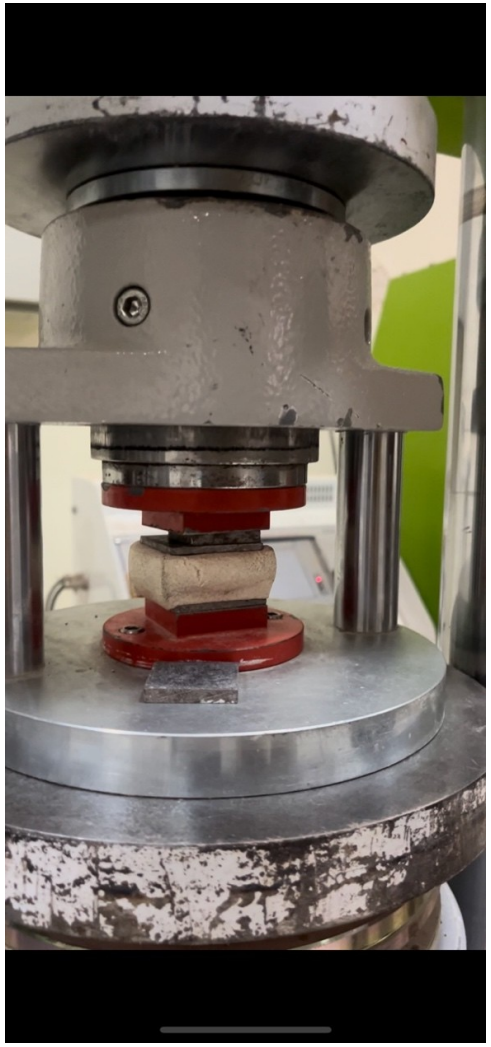
The way to a stronger composite is to take water out, not to put more shell in.

Linear over the range tested, $R^2 = 0.99$ — three formulations, therefore a hypothesis rather than a law. · Fitted relation: σc = 6.93 (solids : water) + 0.04

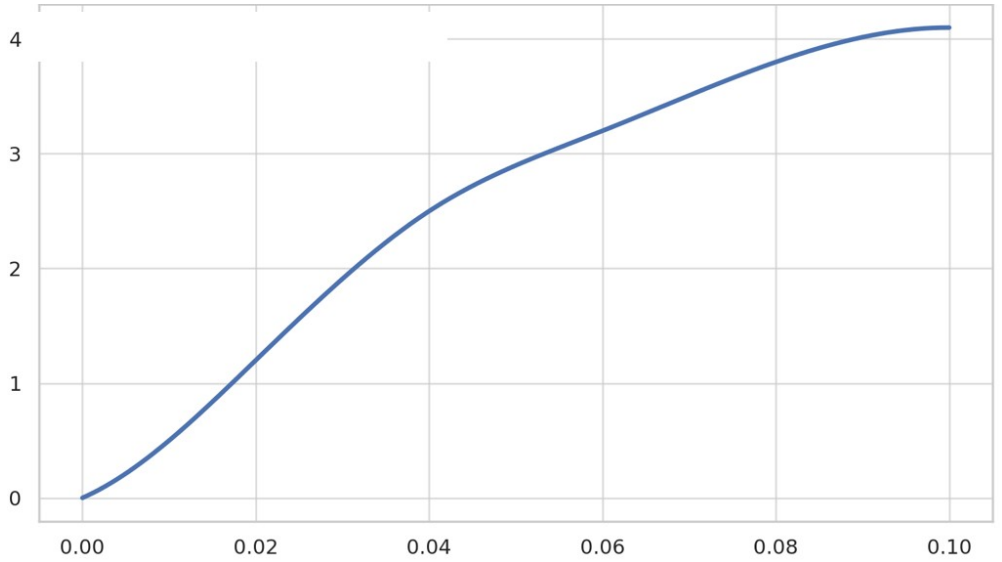


Solids : water ratio shown beneath each formulation identifier.

Compressive failure



Compressive stress (MPa) – strain, C3



Observed response

- Linear to approximately 1 MPa, then progressive loss of stiffness
- Peak stress near 10% strain, with no abrupt load drop
- Continued deformation beyond peak with retained integrity

Failure proceeds by progressive crushing rather than by fracture.

The cube has barrelled: shorter, wider, and still a single body.

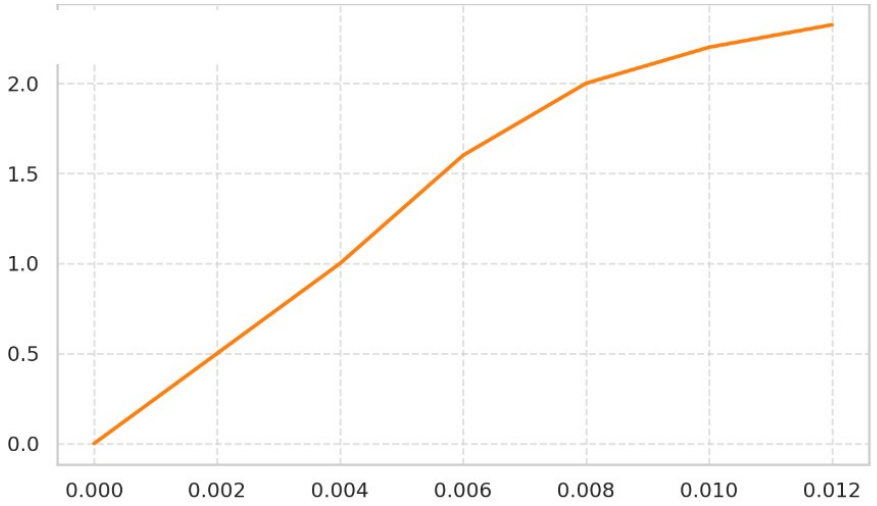
The strain scale assumes ten per cent deformation from observation, not instrumented measurement. · Bulk density 1.38 g/cm^3, mould-volume basis.

Flexure

Measured values — ASTM D790-17, three-point

ID	Glycerol % solids	F_{max} N	σF MPa	EF MPa	ε peak
F1	18.8	—	*not measurable*	—	—
F2	10.3	310	2.325	192	0.0121

Flexural stress (MPa) – strain, F2



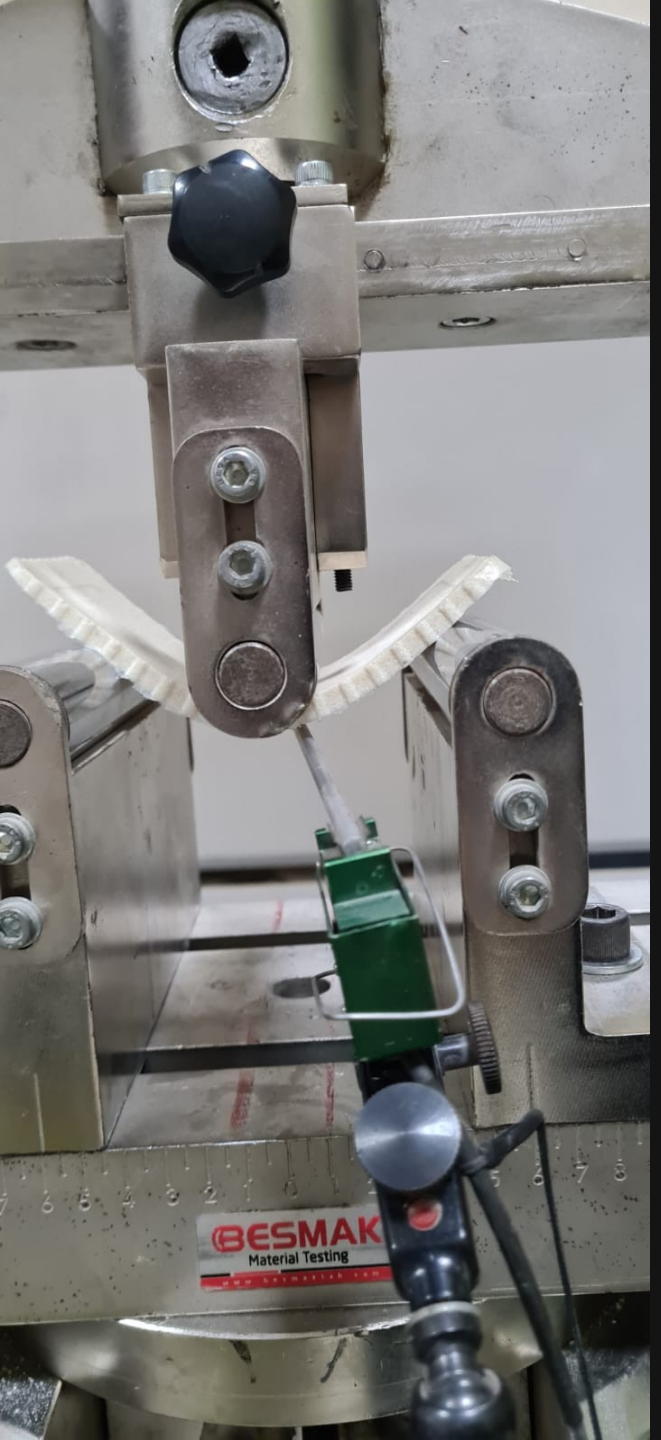
The transition is sharp, and it lies between those two compositions.

Flexural strength computed as $\sigma_F = 3FL / 2bd^2$ on the full rectangular section.

Limitation

The specimen is kerfed on its tension face while the standard assumes a solid rectangle, so the modulus reported here is a lower bound.

Confounded with solids concentration: batch mass was held constant, so water rose as glycerol fell.







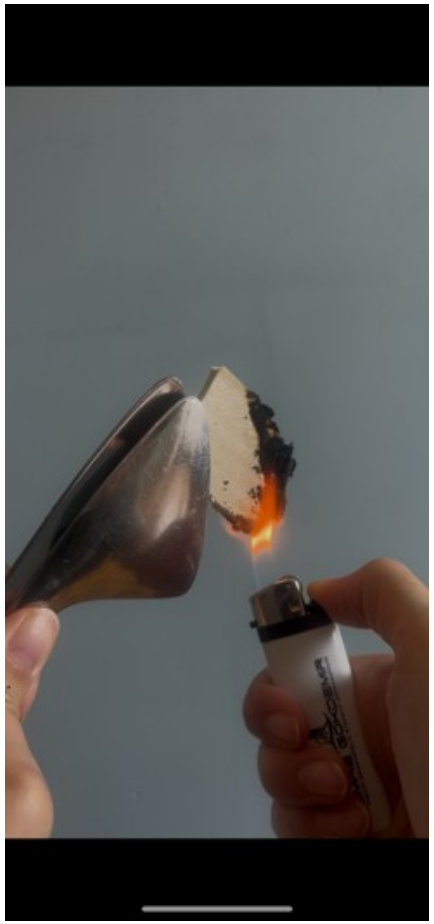
Flexural deformation

Both test programmes gave the same result: substantial deformation with the integrity retained.

The beam is deflected well past the point where a comparable ceramic would have parted, and it is still intact.

Three-point rig at high deflection · kerfed tension face · panel arched by hand

Fire, water, density



Ignitability

≈ 10 s

to ignition under direct flame

4 s

to self-extinguish after flame removal

- Charring remains local
- No lateral flame spread
- No melting and no dripping

It is an ignitability test, and there is no Euroclass classification behind it.

ISO 11925-2:2020. No cone calorimetry, smoke or toxicity data.

Water absorption, 24 h immersion

0.7%

mass gain. It is the coating that limits humid service rather than the composite itself.

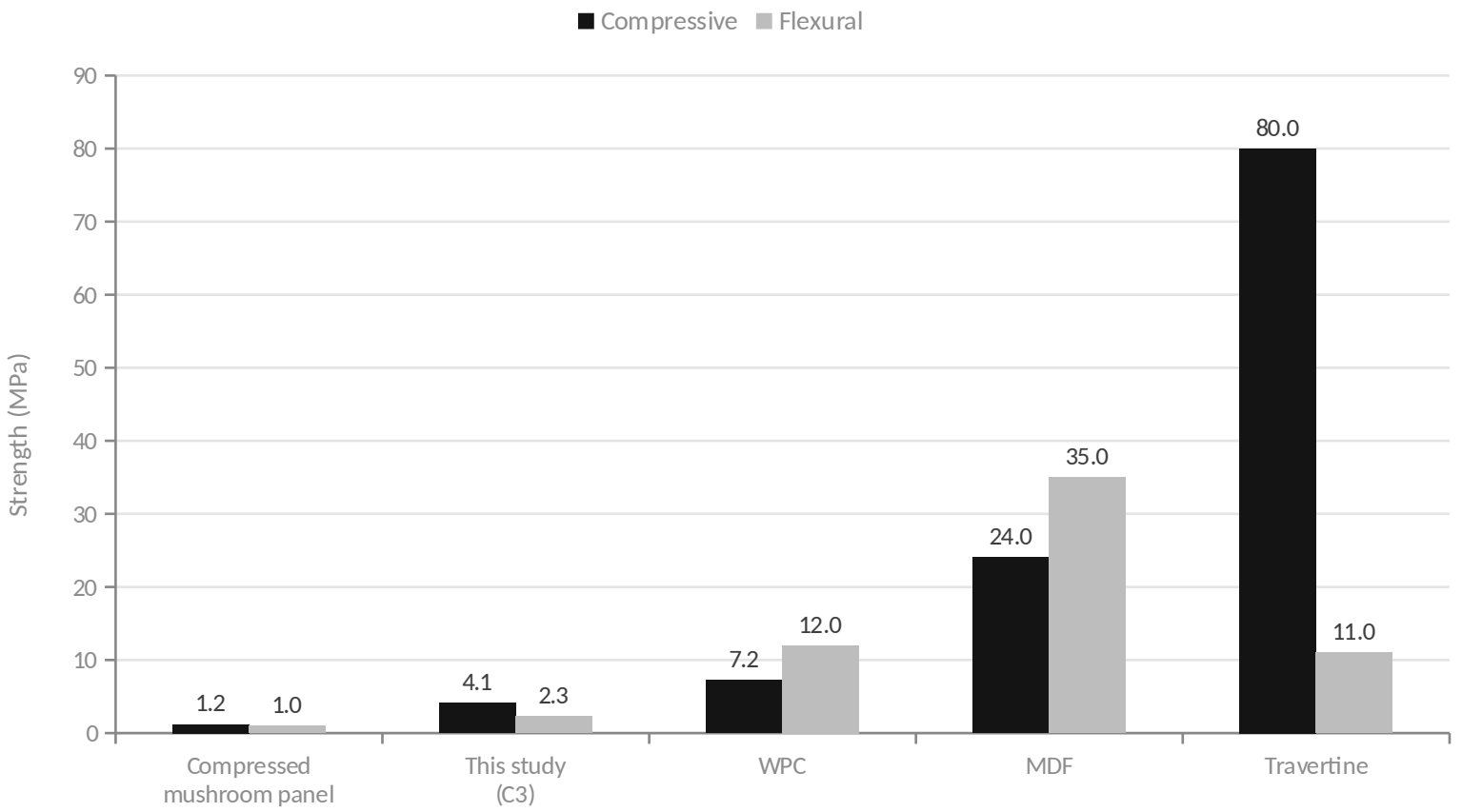
Bulk density

Material	g/cm³	Reference
This study (C3)	1.38	—
Mycelium, cellulose	0.3 – 0.8	Jones et al. (2020)
Fired clay brick	1.6 – 2.0	William & Owen (1995)

Mould-volume basis, therefore a lower bound.

Conventional materials

Compressive and flexural strength against interior surface materials



Velasco & Gaona (2011) · Klyosov (2007) · Nemli & Aydın (2007) · Altun (2006)

On strength alone this composite is not competitive with engineered wood or stone.

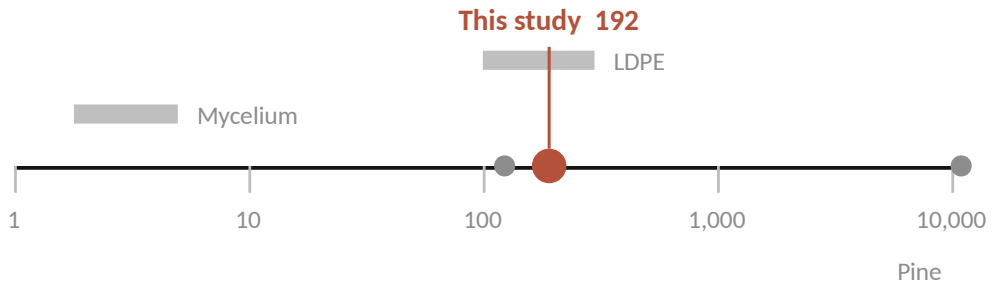
It has one sixth the compressive strength of MDF and one twentieth that of travertine.

Bio-based composites

Compressive strength

Material	σc MPa	Reference
This study (C3)	**4.1**	—
Mycelium composite	0.145 – 0.452	Jones et al. (2020)
Hempcrete	0.3 – 0.6	Arnaud & Gourlay (2012)
Eggshell-filled PLA	2 – 5	Kim & Pal (2010)
Mineral-filled biocomposite	2.1 – 39.6	Shah et al. (2016)

Elastic modulus, MPa, logarithmic



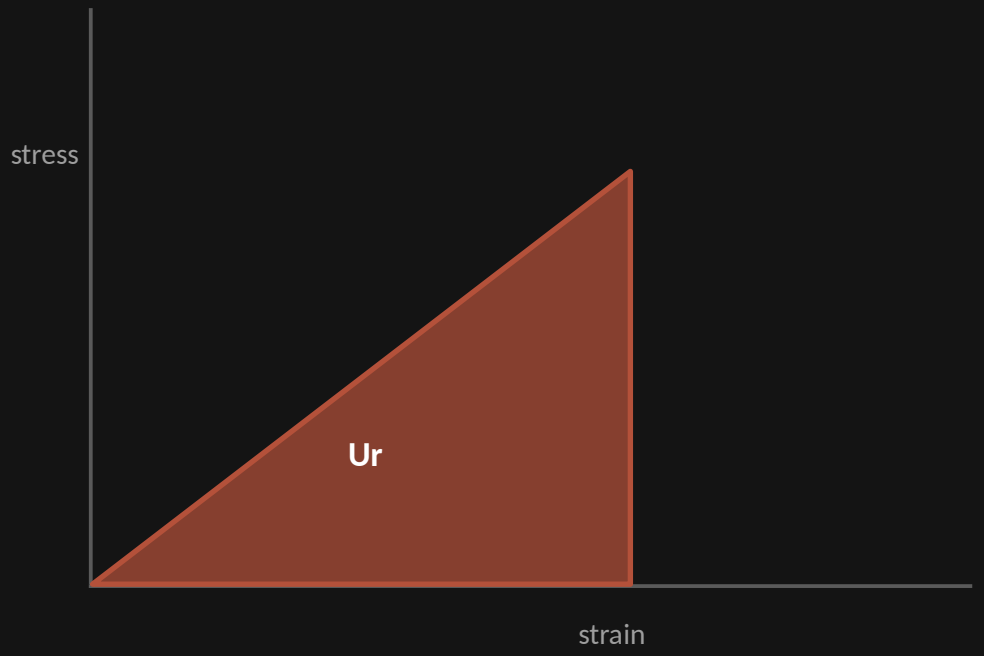
Unlabelled points: soy-based bioplastic 124, flax-fibre PLA 198, pine 11 000.

It was cast from water and dried at room temperature, with no oven, no autoclave and no extruder, and it still reaches the stiffness of melt-processed bio-polymers.

An order of magnitude above the fungal and lime-binder family in compression; wood remains two orders of magnitude above. · Values are reported as published; specimen geometry and test method differ between sources.

Modulus of resilience

Elastic energy absorbed per unit volume



The shaded area is the energy absorbed per unit volume before permanent damage: ½ σ ε, and with ε = σ / E it reduces to the expression at right.

$$Ur = \sigma^2 / 2E$$

14.1 kJ/m³

computed from this study's own σF = 2.325 MPa and EF = 192 MPa

The modulus sits in the denominator.

Strength records the load at which a material stops performing. It does not record whether failure is then graceful or catastrophic.

Limitation

A comparative table cannot be assembled, because the literature reports strength and modulus in separate studies.

Resilience as a property of systems that absorb disturbance and retain function, after Holling (1973).

Application envelope

Modular proposal derived from the measurements



Demountable modules on a reversible fixing, replaceable one unit at a time.

No life-cycle assessment was carried out, so these are qualitative potentials rather than quantified claims.

Conditions of use, each traced to a measured result

Non-loadbearing surfaces

self-weight and handling loads only; σc = 4.1 MPa

Dry interiors

not wet rooms, not façades; absorption depends on coating integrity

Away from ignition sources

self-extinguishing, but unclassified for reaction to fire

Modular rather than sheet

damage is replaced one unit at a time

Every constituent, the shellac included, is biodegradable; reversible fixing recovers the module intact. What is displaced is quarried limestone.

Limitations and further work

Each limitation, and the experiment that resolves it

Limitation	Resolved by
No dispersion data — individual specimen values not retained	**Repeat the series with individual values retained**
Compressive strain assumed from observation	**Instrument the strain; measure the curve**
Shrinkage unquantified, so density is a lower bound	**Record post-drying geometry**
Kerfed section calculated as a solid rectangle	**Test an un-kerfed section**
Flexural series confounded — glycerol and solids moved together	**Vary glycerol at constant solids concentration**
Durability, fire classification and life-cycle assessment untested	Outside this paper; named as further work

Every value reported here is a lower bound.

Conclusions

01	**A protocol that can be repeated**	every production parameter reported
02	**Packing density governs compressive strength**	not mineral content
03	**Plasticiser content sets brittle or ductile response**	the transition is sharp
04	**Bio-polymer stiffness without thermal processing**	water-cast, ambient-dried
05	**Strength on its own is not an adequate criterion**	report σ and E together

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Testing: Advanced Construction Materials Research Laboratory, Hacettepe University, with thanks to Dr Hüseyin İlcan. Derived from the author's doctoral dissertation, Hacettepe University, 2025.

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Standards applied: ASTM D790-17 · ISO 11925-2:2020 · ASTM D570-98(2018).